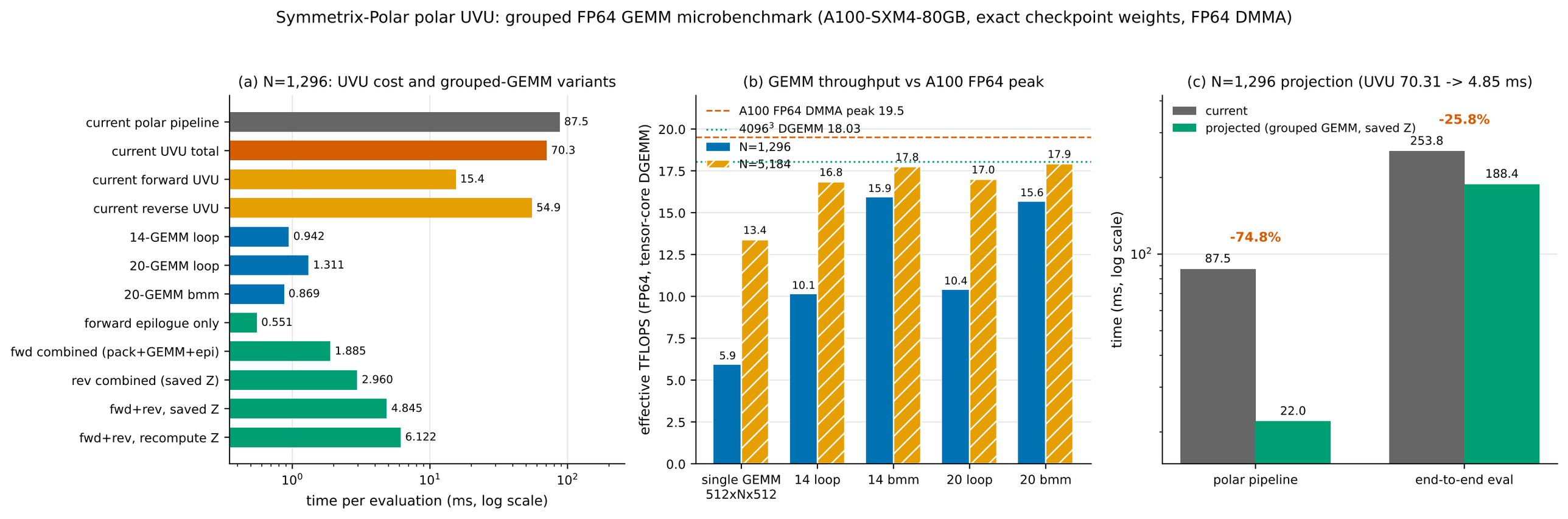
Symmetrix-Polar polar UVU: grouped FP64 GEMM microbenchmark (A100-SXM4-80GB, exact checkpoint weights, FP64 DMMA)
(a) N=1,296: UVU cost and grouped-GEMM variants
current polar pipeline
current UVU total
current forward UVU
current reverse UVU
14-GEMM loop
20-GEMM loop
20-GEMM bmm
forward epilogue only
fwd combined (pack+GEMM+epi)
rev combined (saved Z)
fwd+rev, saved Z
fwd+rev, recompute Z
87.5
70.3
15.4
54.9
0.942
1.311
0.869
0.551
1.885
2.960
4.845
6.122
time per evaluation (ms, log scale)
(b) GEMM throughput vs A100 FP64 peak
A100 FP64 DMMA peak 19.5
4096³ DGEMM 18.03
N=1,296
N=5,184
effective TFLOPS (FP64, tensor-core DGEMM)
single GEMM 512xNx512
14 loop
14 bmm
20 loop
20 bmm
5.9
13.4
10.1
16.8
15.9
17.8
10.4
17.0
15.6
17.9
(c) N=1,296 projection (UVU 70.31 -> 4.85 ms)
current
projected (grouped GEMM, saved Z)
time (ms, log scale)
-74.8%
-25.8%
87.5
22.0
253.8
188.4
polar pipeline
end-to-end eval